Fiscal Year:	FY 2006	**Task Last Updated:**	FY 12/26/2007
PI Name:	Brady, John F. Ph.D.		
Project Title:	Inertial Effects in Suspension Dynamics		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Dynamics and stability		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	11/01/2001	**End Date:**	10/31/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:			
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The rheological constitutive behavior of small sub-micron-sized particle suspensions is relatively well understood, at least for hard-sphere-like particles in Newtonian fluids. In these colloidal dispersions, inertial effects are unimportant and viscous hydrodynamic and thermal forces compete to set the microstructure and determine properties. At the other extreme is a system of very large particles suspended in air. In these granular flows, the fluid plays little role in the dynamics and the particle collisions transport momentum in the same manner as in a dense gas. Granular flows differ from those of a molecular gas in that the particle collisions dissipate energy. Thus the rate of doing work by the flow directly influences the structure and the rheological response. Between these two extremes of small and large particles little is known fundamentally about the rheological behavior of suspension flows.
In this intermediate regime the stress in these flows depends on both the Reynolds number of the fluid, Re, and the Stokes number (or particle Reynolds number), St. Ideally, one would like to vary separately the effects of fluid and particle inertia (i.e., vary separately the Reynolds and Stokes numbers) in order to understand the mechanisms of

Task Description:	rheological response. It is difficult to vary the separate effects because of sedimentation and bulk phase separation, thus pointing to the need for microgravity. This research program is directed at bridging the gap between granular media rheological models and the small Re, small St limit through computer simulation. The objective of this research is to understand the role of fluid inertia in suspension dynamics. The inertia of the particles will be studied separately from the inertia of the fluid in order to isolate the various mechanisms inherent in the rheology of a fluid-particle suspensions. Fluid-solid particle mixtures are encountered in a number of natural and man-made settings, ranging from sediment transport in coastal estuaries to the coating of photographic emulsions. Transporting and processing these multiphase materials requires a knowledge of their response to pressure and shearing forces; that is, their rheological constitutive behavior.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Two areas of work under the general heading of Inertial Effects in Suspension Dynamics have been conducted with the support of this grant. The first was the the effect of particle inertia on the structure and rheology of suspensions. The second was on the effects of particle inertia on the shear-induced diffusive motion of particles. The major conclusions of this study are that there basically exist two distinct states as the effect of particle inertia is varied. At low particle inertia (or low Stokes number) – wet suspensions – viscous forces dominate, stresses scale linearly with the shear rate and the fluctuational motion of the particles as characterized by the suspension or granular temperature is small. This is the so-called ‘quenched’ state. At high particle inertia (or high Stokes number) the fluctuational motion is large, corresponding to a large suspension temperature – the so-called ‘ignited’ state. At high Stokes number – dry suspensions – stresses scale inertially and are proportional to the shear rate squared. The transition from the quenched to the ignited state occurs at a Stokes number of roughly 10 over the wide range of volume fractions studied, from 1% to 45%. A similar change in regime is observed for the diffusive motion. Although the scalings of the shear-induced diffusivity in the two regions of low and high Stokes number are the same – linear in the shear rate – the magnitude of the diffusive motion undergoes an abrupt change at Stokes numbers of order 10.
Bibliography Type:	Description: (Last Updated: 12/26/2007)
Articles in Peer-reviewed Journals	Leshansky A, Brady JF. "Dynamic structure factor study of diffusion in strongly sheared suspensions." Journal of Fluid Mechanics 2005 Mar;527:141-69. http://dx.doi.org/10.1017/S0022112004002903 , Mar-2005
Articles in Peer-reviewed Journals	Sierou A, Brady JF. "Shear-induced self-diffusion in non-colloidal suspensions." Journal of Fluid Mechanics 2004 May;506:285-314. http://dx.doi.org/10.1017/S0022112004008651 , May-2004